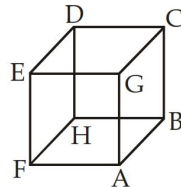


Dice and Cube

Dice

Dice is a cube. In cube there are 6 faces.

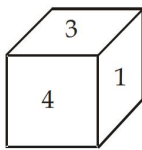


1. There are 6 faces in the cube - ABCG, GCDE, DEFH, BCDH, AGEF and ABHF.
2. Always four faces are adjacent to one face
3. CDEG is the upper face of the cube.
4. ABHF is the bottom face of the cube.

✦ Types of Dice

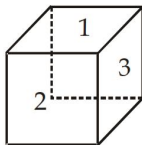
1. Simple Dice
- (2) Standard Dice

1. **Simple Dice:** In simple dice, sum of the opposite side are not seven. It means sum of any two adjacent sides is equal to seven.



Here, $3 + 4 = 7$, So it is simple dice.

2. **Standard dice:** In standard dice, the sum of two opposite sides is seven. In other words the sum of any two adjacent is not seven.



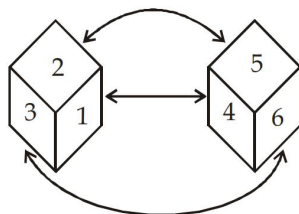
From the figure;
 $1 + 2 = 3$, $2 + 3 = 5$, $3 + 1 = 4$ does not have total seven. So, we can say that the opposite sides no. must be equal to be seven.

In Standard dice:

- 1 will be opposite to 6
- 2 will be opposite to 5
- 3 will be opposite to 4

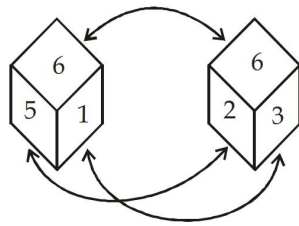
Some Rules are:

1. When two different position of dice have different number then they are opposite to each other.



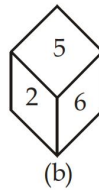
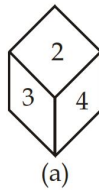
3 will be opposite to 4
 1 will be opposite to 6
 5 will be opposite to 2

2. If one side of given two dice is common in the same position then remaining are opposite to each other.

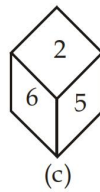


1 will be opposite to 3
5 will be opposite to 2
6 will be opposite to 4.

- If common sides are not in same position, then we rotate it clockwise or anti-clockwise to bring it into the same position.

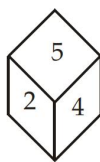
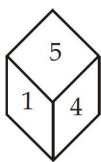


Here, common side is 2, but not in the same position, so we rotate (b) in clockwise direction.



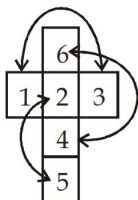
Now, 2 Will be opposite to 1
4 will be opposite to 5
3 will be opposite to 6

3. If two sides of a given dice are common in the same position, then remaining is opposite to each other.



1 will be opposite to 2.

✦ Expanded form of Dice



6 will be opposite to 4
1 will be opposite to 3
2 will be opposite to 5

Cube

Cubes are 3-dimensional objects, they also follow same property as dice rather than numbering. Cube have all side same and cuboid have different length, height and breadth. They also have 8 vertex, 6 faces and 12 edges. In exam, these topics contain cutting of both cube and cuboid and counting the number of small cubes and cuboid.

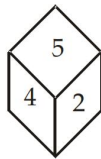
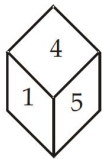
- * If a cube is painted on all of its surfaces and divided into smaller cubes of equal size, then after separation number of smaller cubes so obtained will be calculated as :

*
$$n = \left[\frac{\text{Side of larger cube}}{\text{Side of smaller cube}} \right] \quad (n = \text{number of smaller cubes along each edge})$$

1. Total number of smaller cubes = n^3
2. 3 faces coloured = 8
3. 2 faces coloured = $12(n - 2)$
4. 1 face coloured = $6(n - 2)^2$
5. No face coloured = $(n - 2)^3$
6. Total number of coloured cube = $n^3 - (n - 2)^3$
7. Number of cubes with atleast 2 face coloured = $12(n - 2) + 8$

Solved Examples

1. Below two positions of a dice are shown. What will be on opposite of the face at which 1 is written?

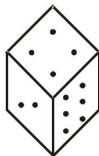


- (a) 2 (b) 3
(c) 4 (d) 1

Sol. The digits that can not be opposite to 1 are 4 and 5. The possible numbers opposite to 1 are 2 and 3 and 6, which comes in alternative, first will be the answer.

1 is opposite to 2

2. There are three different positions of a dice shown below. How many dots lie opposite to 2 dots?



(i)



(ii)



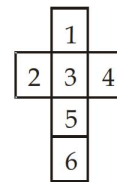
(iii)

- (a) 1 (b) 3
(c) 5 (d) 6

Sol. From figure (ii) and (iii), we conclude that 1, 6, 3 and 4 dots lie adjacent to 5 dots.

Therefore, 2 dots must lie opposite to 5 dots. conversely, 5 dots must lie opposite to 2 dots.

3. What will be opposite to 5 and 4?



Sol. 1 is opposite to 5
2 is opposite to 4
3 is opposite to 6

4. A 6 cm cube is cut into 2 cm small cube. How many small cubes can be obtained from this?

- (a) 108 (b) 156
(c) 27 (d) 64

Sol. Here, $n = \frac{6}{2} = 3$

$\therefore$ Number of smaller cube = $n^3 = 3^3 = 27$

5. A cube is coloured red on all of its faces. It is then cut into 27 smaller cubes of equal size. The smaller cubes so obtained are now separated.

- (1) How many smaller cubes will have two surfaces painted with red colour and remaining side without any colour?
- (2) How many smaller cubes have only three surface painted with red colour?
- (3) How many smaller cubes have only one surface painted with red colour?
- (4) How many smaller cubes have no surface coloured?

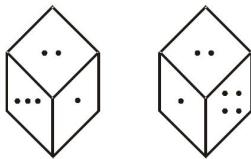
Sol. $n = \sqrt[3]{27} = 3$

- (1) No. of smaller cubes painted in two sides
 $= 12(n - 2) = 12(3 - 2) = 12$
- (2) No of smaller cubes with three surface
 painted with red surface = 8

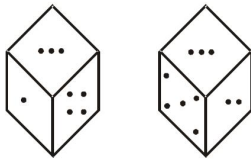
- (3) No of smaller cubes with one surface painted
 with red surface
 $= (n - 2)^2 \times 6 = (3 - 2)^2 \times 6 = 6$
- (4) Number of smaller cubes with no surface
 painted with red colour
 $= (n - 2)^3 = (3 - 2)^3 = 1.$

Practice Set

1. On the basis of two figures of dice. You have to tell what number will be there on the opposite side of number 2?

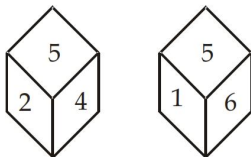


- (a) 3
 (c) 1
- (b) 6 or 5
 (d) cannot be determined
2. On the basis of two figures of dice you have to tell what number will be there on the opposite side of number 1?



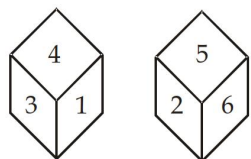
- (a) 3
 (c) 6
- (b) 4
 (d) 5

Directions (3-5): On the basis of two figures of dice. Answer the following questions.

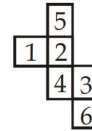


3. Which number is at the opposite of number 2?
 (a) 1
 (c) 3
- (b) 2
 (d) 5
4. Which number is at the opposite of number 4?
 (a) 1
 (c) 3
- (b) 6
 (d) 4
5. Which number is at the opposite of number 5?
 (a) 2
 (c) 3
- (b) 4
 (d) 6

Directions (6-8): Questions given below are based on two figures of dice.

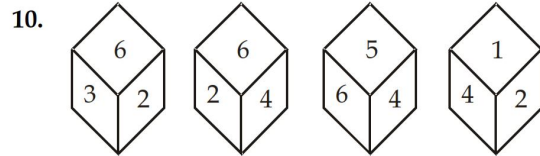


6. Which number is at the opposite of number 4?
 (a) 5
 (c) 6
- (b) 2
 (d) none of these
7. Which number is at the opposite of number 3?
 (a) 2
 (c) 6
- (b) 5
 (d) 4
8. Which number is at the opposite of number 1?
 (a) 2
 (c) 4
- (b) 5
 (d) 6
9. When the following figure is folded to form a cube, then which number is at the opposite of number 1?



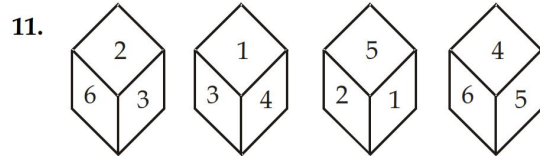
- (a) 3
 (c) 4
- (b) 6
 (d) 5

Directions (10-16): In each of the following questions, four positions of the same dice has been shown. You have to see these figure and select the number/symbol opposite to number/symbol as asked in each of question.



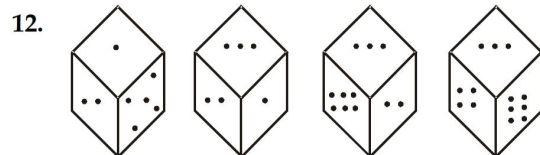
Which number is at opposite of number 6?

- (a) 1
 (c) 3
- (b) 2
 (d) 4



Which number is at the opposite of number 3?

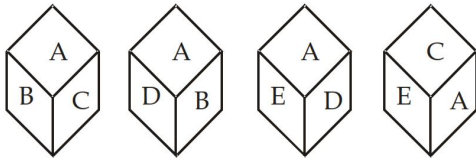
- (a) 5
 (c) 2
- (b) 1
 (d) 6



Which number is at the opposite of number 4?

- (a) 1
 (c) 5
- (b) 2
 (d) 3

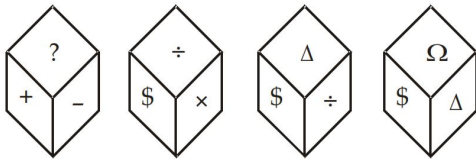
13.



Which alphabet is at the opposite of alphabet E?

- (a) C (b) B
(c) D (d) A

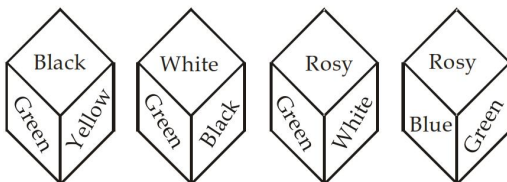
14.



Which symbol is at the opposite of symbol ÷?

- (a) ? (b) \$
(c) x (d) +

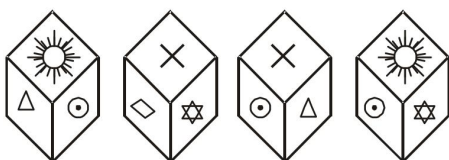
15.



Which colour is at the opposite of the black colour?

- (a) Blue (b) Rosy
(c) Green (d) White

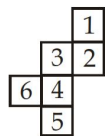
16.



Which symbol is at the opposite of the symbol ?

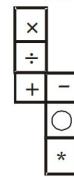
- (a)  (b) 
(c)  (d) 

17. When the following figure is folded to form a cube, then which number is at the opposite of number 2?



- (a) 6 (b) 4
(c) 5 (d) 1

18. When the following figure is folded to form a cube, then which symbol is opposite to ÷ ?



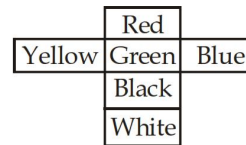
- (a)  (b) 
(c)  (d) 

19. When the following figure is folded to form a cube, then which symbol is the opposite of symbol Δ?



- (a)  (b) 
(c)  (d) 

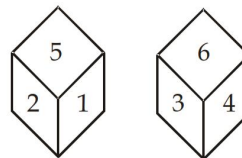
20. When the following figure is folded to form a cube, then which colour is at the opposite of colour white?

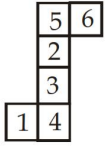
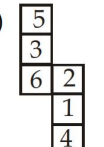
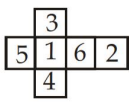
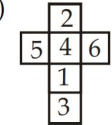


- (a) Green (b) Yellow
(c) Blue (d) Red

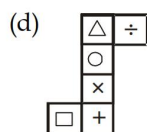
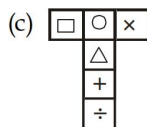
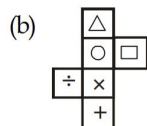
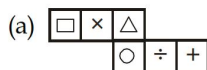
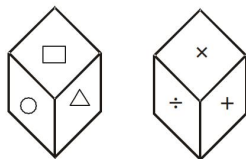
Direction (21 to 22) : In each of the following questions two positions of the same dice have been shown. If from these dice a figure is made then which option will be suitable for answer.

21.

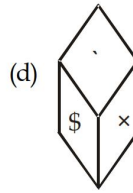
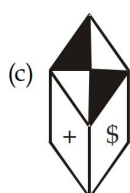
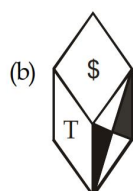
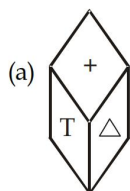
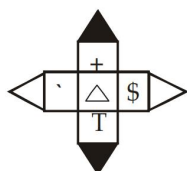


- (a)  (b) 
(c)  (d) 

22.



23. Given below a figure. If we fold the figure and give it shape of a dice then which option will be suitable for answer



24. A cube is coloured red on all of its faces. It is then cut into 125 smaller cubes of equal size. Then how many smaller cubes have three surfaces painted with red colour?
- (a) 4 (b) 8
(c) 12 (d) 10
25. A cube is coloured red on all of its faces. It is then cut into 27 smaller cubes of equal size. The smaller cubes so obtained are now separated. Then how many smaller cubes have no surface coloured?
- (a) 0 (b) 1
(c) 3 (d) 4

26. A cube of 6cm length coloured yellow all of its faces. Then it cut 1 cm length cubes. How many smaller cubes have two surfaces painted with yellow colour?
- (a) 50 (b) 54
(c) 48 (d) 52

Directions (27-30): There are 343 same size of wood cubes from all these cubes a solid cube was made then it coloured red on all of its faces then answer the following question.

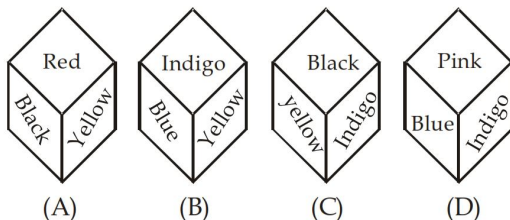
27. How many smaller cubes have only three surfaces painted with red colour?
- (a) 8 (b) 4
(c) 2 (d) 1
28. How many smaller cubes have two surfaces painted with with red colour?
- (a) 40 (b) 60
(c) 44 (d) 80
29. How many smaller cubes have only one surface painted with red colour?
- (a) 200 (b) 175
(c) 150 (d) 125
30. How many smaller cubes have no surface coloured?
- (a) 129 (b) 127
(c) 126 (d) 125

Directions (31-35): A cube of 5 cm has been painted on its surface in such a way that two opposite surface have been painted red colour and other two opposite surface have been painted blue colour. Two remaining surface have been left unpainted. Now the cube is cut into smaller cubes of side 1 cm each.

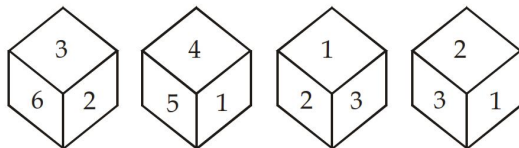
31. How many smaller cubes will have none of the sides painted?
- (a) 10 (b) 15
(c) 9 (d) 45
32. How many cubes will have only two surfaces painted with red and blue colours respectively?
- (a) 15 (b) 20
(c) 25 (d) 17
33. How many cubes will have three surfaces coloured?
- (a) 0 (b) 4
(c) 8 (d) 5
34. How many cubes will have only one surface painted with red and no colour on remaining surfaces?
- (a) 50 (b) 40
(c) 30 (d) 60
35. If cubes having only red colour are removed then how many cubes will left?
- (a) 100 (b) 110
(c) 105 (d) 75

Directions (36-41): A cube has been painted on its surfaces in such a way that two opposite surfaces have been painted black and white. other two opposite surfaces have been painted green and rosy. One surface is blue and one surface is purple. It is then cut into 216 smaller cubes of equal size.

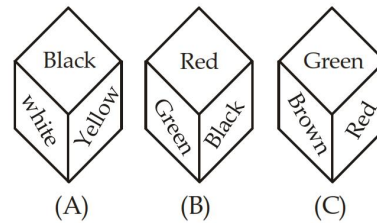
36. How many cubes will have only two surfaces painted green and white and remaining side without colour?
 (a) 12 (b) 16
 (c) 20 (d) 4
37. How many cubes will have atleast blue colour on its surface and remaining sides without any colour?
 (a) 16 (b) 20
 (c) 8 (d) 10
38. How many such cubes are there which one surface is rosy and one surface is either blue or purple and other surface there is no colour?
 (a) 8 (b) 12
 (c) 10 (d) 25
39. How many cubes will have only two surfaces painted with blue and black, respectively and there is no colour of other surface?
 (a) 4 (b) 8
 (c) 12 (d) 64
40. How many cubes will have none of the sides painted?
 (a) 16 (b) 20
 (c) 64 (d) 28
41. How many cubes will have three surfaces coloured?
 (a) 8 (b) 16
 (c) 32 (d) None of these
42. Four different positions of a dice have been shown in the following figure. What will be the colour of the surface opposite to red?



- (a) Indigo (b) Blue
 (c) Pink (d) Either blue or pink
43. Four positions of a dice are given below. Identify the number at the bottom when top's number is 6:



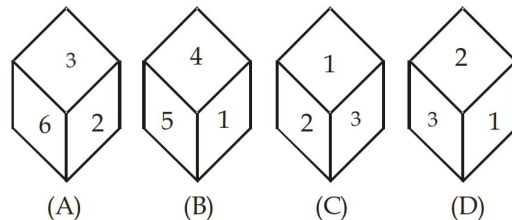
- (a) 5 (b) 1
 (c) 3 (d) 4
44. Three different positions of the same dice have been shown in the figure. Which of the following colours will be opposite to the red surface?



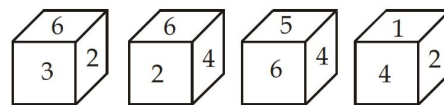
- (a) White (b) Yellow
 (c) Brown (d) Green

Distinct Questions

45. Four positions of a dice are given in the question figures. Identify the number at the bottom when top is 6.



- (a) 1 (b) 4
 (c) 5 (d) Can't say
46. Consider the following pictures of a dice:



What is the number opposite 1?

- (a) 2 (b) 3
 (c) 5 (d) 6

Directions (47-60): Read the following information and answer the questions given below:

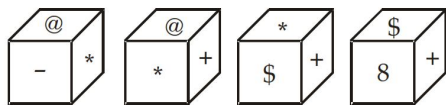
A solid cube of each side 15 cm has been painted green, blue and yellow on pairs of opposite faces. It is then cut into cubical blocks of each side 3 cm.

47. How many cubes have no face painted?
 (a) 27 (b) 54
 (c) 36 (d) 50
48. How many cubes have only one face painted?
 (a) 27 (b) 54
 (c) 36 (d) 50
49. How many cubes have only two faces painted?
 (a) 27 (b) 54
 (c) 36 (d) 50
50. How many cubes have only three faces painted?
 (a) 16 (b) 18
 (c) 12 (d) 8

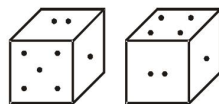
51. How many cubes have three faces painted with different colours?
 (a) 18 (b) 4
 (c) 16 (d) 8
52. How many cubes have two faces painted green and blue and all other faces unpainted?
 (a) 12 (b) 16
 (c) 18 (d) 8
53. How many cubes have two faces painted Blue and yellow and all other faces unpainted?
 (a) 16 (b) 12
 (c) 18 (d) 8
54. How many cubes have two faces painted yellow and green and all other faces unpainted?
 (a) 12 (b) 16
 (c) 18 (d) 8
55. How many cubes have one face painted blue and other faces unpainted?
 (a) 12 (b) 16
 (c) 18 (d) 8
56. How many cubes have one face painted yellow and other faces unpainted?
 (a) 12 (b) 16
 (c) 18 (d) 8
57. How many cubes have one face painted green and other faces unpainted?
 (a) 12 (b) 16
 (c) 18 (d) 8
58. How many cubes have atleast one face blue?
 (a) 27 (b) 54
 (c) 36 (d) 50
59. How many cubes have atleast one face green?
 (a) 27 (b) 54
 (c) 36 (d) 50
60. How many cubes have atleast one face yellow?
 (a) 27 (b) 54
 (c) 50 (d) 36

Previous year Questions (Memory Based)

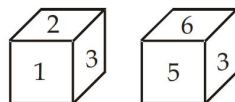
1. Which symbol will be on the face opposite to the face with symbol *?



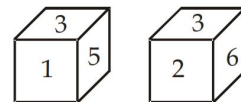
- (a) @ (b) \$
 (c) 8 (d) +
2. Two positions of dice are shown below. How many points will appear on the opposite to the face containing 5 points?



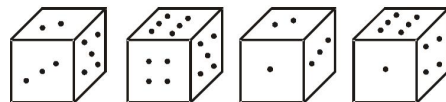
- (a) 3 (b) 1
 (c) 2 (d) 4
3. Which digit will appear on the face opposite to the face with number 4?



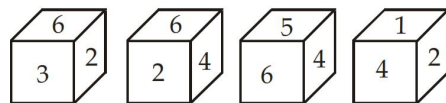
- (a) 3 (b) 5
 (c) 6 (d) 2/3
4. Two positions of a dice are shown below. Which number will appear on the face opposite to the face of the number 5?



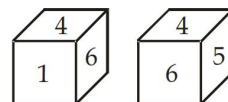
- (a) 2/6 (b) 2
 (c) 6 (d) 4
5. How many points will be on the face opposite to in face which contains 2 points?



- (a) 1 (b) 5
 (c) 4 (d) 6
6. Which number is on the face opposite to 6?

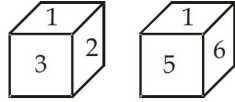


- (a) 4 (b) 1
 (c) 2 (d) 3
7. Two positions of a dice are shown below. When number '1' is on the top. What number will be at the bottom?

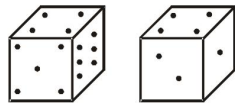


- (a) 3 (b) 5
 (c) 2 (d) 6

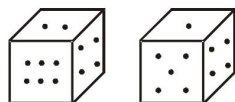
8. Two positions of a cube with its surfaces numbered are shown below. When the surface 4 touch the bottom, what surface will be on the top?



- (a) 1
(b) 2
(c) 5
(d) 6
9. Here two positions of dice are shown. If there are two dots in the bottom, then how many dots will be on the top?



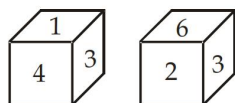
- (a) 2
(b) 3
(c) 5
(d) 6
10. Two positions of dice are shown below. How many points will be on the top when 2 points are at the bottom?



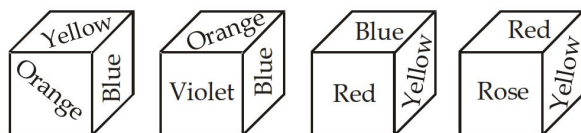
- (a) 6
(b) 5
(c) 4
(d) 1
11. Here 4 positions of a cube are shown. Which sign will be opposite to '+'?



- (a) %
(b) -
(c) x
(d) \$
12. Two positions of a cubical block are shown. When 5 is at the top which number will be at bottom?

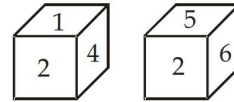


- (a) 1
(b) 2
(c) 3
(d) 4
13. From the four positions of a dice given below, find the color which is opposite to yellow?

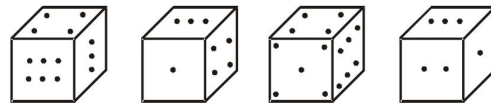


- (a) Violet
(b) Red
(c) Rose
(d) Blue

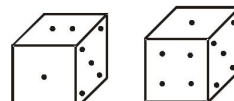
14. When the digit 5 is on the bottom then which number will be on its upper surface?



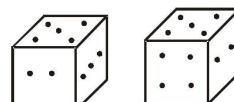
- (a) 1
(b) 3
(c) 4
(d) 6
15. How many points will be on the face opposite to the face which contain 3 points?



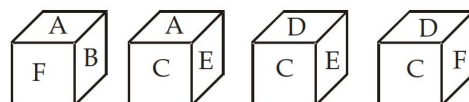
- (a) 2
(b) 4
(c) 5
(d) 6
16. observe the dots on the dice (one to six dots) in the following figures. How many dots are contained on the face opposite to the containing four dots?



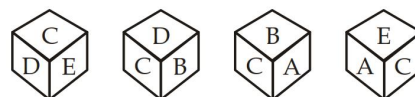
- (a) 2
(b) 3
(c) 5
(d) 6
17. Two positions of a dice are shown below. When 3 points are at the bottom, how many points will be at the top?



- (a) 2
(b) 5
(c) 4
(d) 6
18. From the positions of a cube are shown below, which letter will be on the face opposite to face with 'A'?



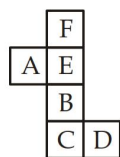
- (a) D
(b) B
(c) C
(d) F
19. Four position of dice are given below. Which letter will be opposite to D.



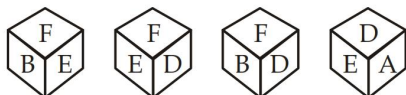
- (a) B
(b) C
(c) D
(d) A

20. Which of the following cubes can be created by folding the given figure.

Question figure :



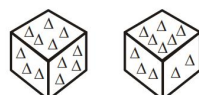
Answer figure :



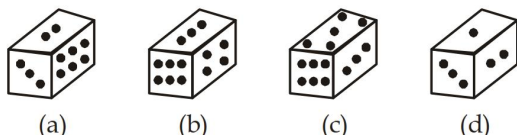
21. Two positions of dice are given. Which number would be at the top when bottom is 2.



- (a) 4
(c) 5
22. Two positions of a dice are given. How many triangles would be at the top when bottom is one triangle present.

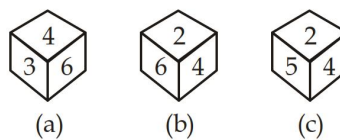


- (a) 4
(c) 2
23. Four positions of a dice are given which spot opposite to face of dice having 6 spots.



- (a) 1
(c) 5
(b) 3
(d) 2

24. Given various views of a dice, what number will be at the top if its bottom digit 4 is available.



- (a) 1
(c) 5
(b) 6
(d) 2

Direction (25-30): If any cube which opposite side are coloured with Red, Green and Blue, and its side is 15 cm. If it is cut into smaller cubes of side 5 cm, then

25. Find out number of cubes that are coloured with only green colour.

- (a) 1
(c) 3
(b) 2
(d) 4

26. Find out number of cubes which coloured with at least two colours.

- (a) 15
(c) 20
(b) 18
(d) 29

27. The numbers of cubes which doesn't have any colour

- (a) 1
(c) 14
(b) 15
(d) 18

28. The number of cubes that have red colour.

- (a) 1
(c) 14
(b) 15
(d) 18

29. The number of cubes that have blue but not with red.

- (a) 6
(c) 10
(b) 8
(d) 12

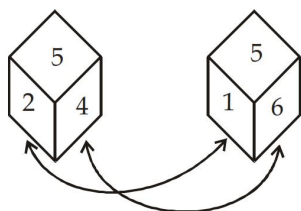
30. The number of cube that have only one face coloured.

- (a) 16
(c) 6
(b) 10
(d) 8

Practice Set Solutions

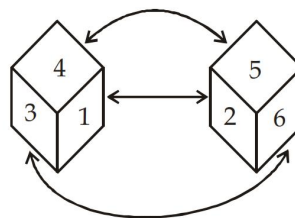
1. (b); From both figures it is clear that numbers adjacent to 2 are 3, 1 and 4. Hence number 5 or 6 will be opposite to number 2.
2. (d); Clearly that position is rotated from left to right twice. Hence number 5 will be opposite to number 1.

(3 to 5)

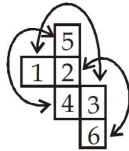


3. (a); Clearly number 1 will be opposite to number 2.
4. (b); Clearly number 6 will be opposite to number 4.
5. (c); From both figure, it is clear that numbers adjacent to 5 are 2, 4, 1 and 6. Hence number 3 will be opposite to number 5.

(6 to 8)



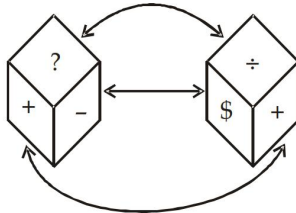
6. (a); Clearly number 5 will be opposite of number 4.
 7. (c); Clearly number 6 will be opposite of number 3.
 8. (a); Clearly number 2 will be opposite of number 1.
 9. (a); In such kind of figure the opposite sides of dice will be in such a way showing below.



Clearly number 3 will be opposite to number 1.

10. (a); From the figure (1), (2) and (3) it is clear that numbers adjacent to 6 are 2, 3, 4, 5. Hence number 1 will be opposite to number 6.
 11. (a); From the figure (1) and (2) It is clear that numbers adjacent to 3 are 2, 6, 1 and 5. Hence number 4 will be opposite to number 3.
 12. (b); If we rotate position (3) from left to right, then we come to know that number 2 will be opposite to number 4.
 13. (b); If we rotate position (1), (2) and (3) from left to right respectively then we come to know letter E will be opposite to letter B.

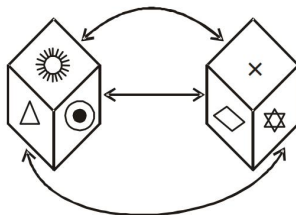
14. (a);



Symbol ? will be opposite to symbol ÷

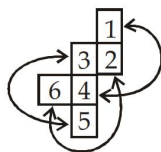
15. (b); If we rotate position (1), (2) and (3) from up to down respectively then we come to know, red will be opposite to black.

16. (a);



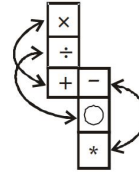
x is opposite to the symbol

17. (a); If we shape the figure in a dice then opposite sides will be as shown below.



Clearly, number 6 will be opposite to number 2.

18. (b); If we shape the figure in a dice then opposite sides will be as shown below.



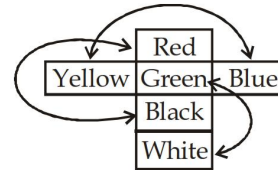
Clearly, symbol 'O' opposite to symbol ÷.

19. (c); If we shape the figure in a dice then opposite side will be as shown below.



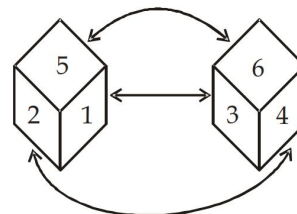
Clearly, symbol $\star$ is opposite of symbol Δ .

20. (a); If we shape the figure in a dice then opposite sides will be as shown below.

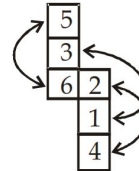


Green colour is opposite to white colour.

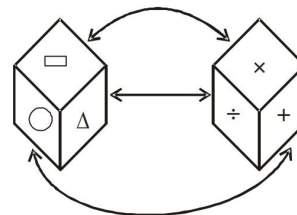
21. (b);



5 will be opposite to 6.
 2 will be opposite to 4.
 1 will be opposite to 3.



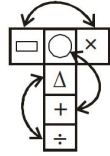
22. (c);



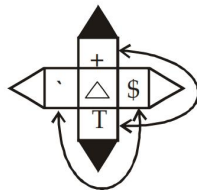
'□' is opposite to '×'

'○' is opposite to '+'

'Δ' is opposite to '÷'



23. (b); If we shape the figure into dice then opposite sides will be shown below.



symbol '+' is opposite to 'T'

Symbol 'Δ' is opposite to '\$'

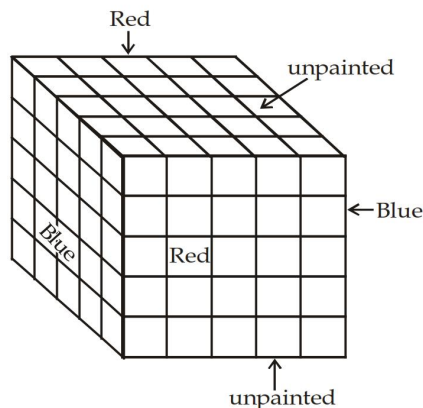
symbol 'Δ' is opposite to 'X'



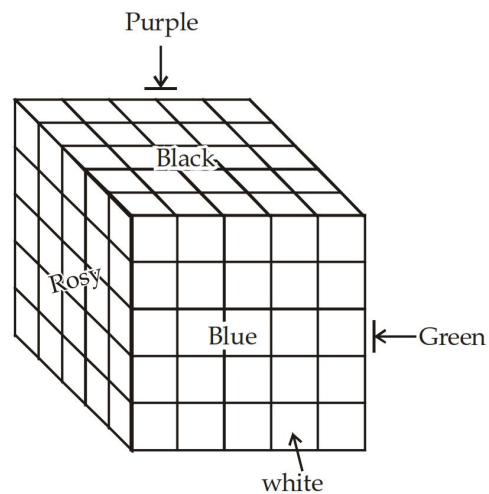
Except (2) in all options opposite symbol are shown at the adjacent surfaces.

24. (b); We know that number of smaller cubes with three surfaces painted is always 8.
25. (b); Number of smaller cubes with no surface painted = $(n-2)^3$. Here $n = \sqrt[3]{27} = 3$
So, required number = $(3-2)^3 = 1$
26. (c); Number of smaller cubes with two surfaces painted = $(n-2) \times 12$
Here, $n = \sqrt[3]{216} = 6$
So, required number = $(6-2) \times 12 = 4 \times 12 = 48$
27. (a)
28. (b); Here, $n = 7$
So, required no. = $(7-2) \times 12 = 5 \times 12 = 60$
29. (c); Number of smaller cubes with one surface painted
 $(n-2)^2 \times 6 = (7-2)^2 \times 6 = (5)^2 \times 6 = 25 \times 6 = 150$
30. (d); Number of smaller cubes with no surface painted = $(n-2)^3 = (7-2)^3 = (5)^3 = 125$

(31 to 35)



31. (d); There are 80 cubes painted at least one side. So smaller cubes whose none side is painted are $125 - 80 = 45$.
32. (b); Clearly, 20 cubes will have only two surfaces painted with red and blue colours respectively.
33. (a); Clearly, No such cubes are there coloured with three surfaces because two opposite side of cubes are unpainted.
34. (c);
35. (d); No. of red colour cubes = 50
No. of remaining cubes = $125 - 50 = 75$
- (36 to 41)



36. (d); Only 4 cubes will have two surfaces painted green and white and remaining side without colour.
37. (a); Clearly 16 cubes will have atleast blue colour on its surface, remaining sides without colour.
38. (a); Clearly 8 cubes there whose one surface is rosy and one surface is either blue or purple.
39. (a); Clearly only 4 cubes are there.
40. (c); Number of smaller cubes with no surface painted = $(n-2)^3$.
Here, $n = \sqrt[3]{216} = 6$
 $\therefore$ Required no = $(6-2)^3 = (4)^3 = 64$
41. (a);
42. (a); 'Red' appears in figure (A) along with 'Black' and 'Yellow'. Also 'Black' and 'yellow' appear together in one more figure (C), along with 'indigo'. Hence, 'Indigo' must be opposite 'Red'.
43. (b); The numbers 2, 3, 4 and 5 cannot be on the face opposite to 1. Therefore, 6 lies opposite to 1.
44. (a); From figure (C) we reject (3) and (4) because green and brown surfaces are adjacent surfaces. After a close look at fig (A) and (C) we come to know that red surface is opposite the white surface.

Distinct Solutions

45. (a); 6 appears in one figure (A) along with 3 and 2. 2 and 3 appear together in another figures (C) and (D) along with 1. Hence, 1 must be opposite to 6.
46. (d); 1 appears in one figure along with 4 and 2. 4 and 2 appear together in another figure along with 6. Hence, 6 must be opposite 1.

(47 to 60):

$$n = \frac{15}{3} = 5$$

Total number of cubes = $5^3 = 125$

Cubes having only one face green = $2(n-2)^2 = 2(5-2)^2 = 18$.

Cubes having one face blue = $2(n-2)^2 = 2 \times 3 \times 3 = 18$

Cubes having one face yellow

$$= 2(n-2)^2 = 2 \times 3 \times 3 = 18$$

Cubes having only one face painted = $18 + 18 + 18 = 54$

Cubes of only two faces painted comprises three types of cube

1st type: one face blue and one face yellow

$$4(5-2) \Rightarrow 4(n-2) = 4 \times 3 = 12$$

2nd type: one face yellow and one face green

$$4(n-2) \Rightarrow 4(5-2) = 4 \times 3 = 12$$

3rd type: one face green and one face blue

$$4(n-2) \Rightarrow 4(5-2) = 4 \times 3 = 12$$

Cubes having only two faces painted = $12 + 12 + 12 = 36$

Cubes which have no face painted = $(n-2)^3 = (5-2)^3 = 3^3 = 27$.

- | | | | |
|---------|---------|---------|---------|
| 47. (a) | 48. (b) | 49. (c) | 50. (d) |
| 51. (d) | 52. (a) | 53. (b) | 54. (a) |
| 55. (c) | 56. (c) | 57. (c) | |
58. (d); Cubes having one face painted blue are 18. Cubes having one face blue and one face green are 12. Cubes having one face blue and one face yellow are 12.
- Cubes having painted with three colours are 8. Hence total such cubes = $18 + 12 + 12 + 8 = 50$.
59. (d);
60. (c);

Previous year Solutions (Memory Based)

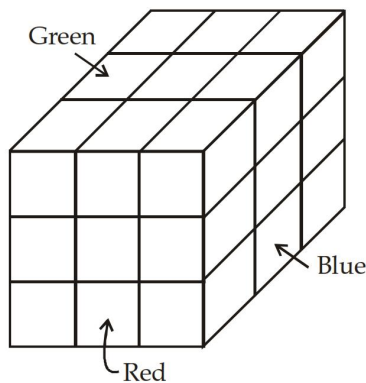
- (c); The symbol of the adjacent faces to the face with symbol * are @, -, + and \$. Hence the required symbol is 8.
- (d); In these two positions one of the common face having 1 point is in the same position. There will be 4 posints on the required face.
- (a); Here the common faces with number 3, are in same positions. Hence 6 is opposite to 2 and 5 is opposite to 1. Therefore 4 is opposite to 3.
- (c); Common faces with number 3, are in same positions. Hence the number of the opposite face to face as number 5 will be 6.
- (d); In first two positions of dice one common face containing 5 is same. The face opposite to the face which contains 2 point, will contains 6 points.
- (b); As the numbers 2, 3, 4 and 5 are adjacent to 6. Hence the number on the face opposite to 6 is 1.
- (b); when 'one' is at the top, then 5 will be at the bottom.
- (a); In these 2 positions one common face with number 1 is in the same position, 2 is opposite 6 and 3 is opposite to 5. therefore opposite to 4 is 1.
- (c); Here the common faces with 4 dots are in same positions. Hence 2 will be opposite to 5.
- (d); Here the common faces with 4 dots are in same positions. Hence 2 will be opposite to 1.
- (c); From position I and III common face with % is in the same position. Hence × is oppositie to +.
- (c); In these 2 positions one common face with number 3, is in same position. Hence 1 is opposite to 6 and 4 and 4 is opposite to 2. Therefore 5 is opposite to 3.
- (a); The colours adjacent to yellow are orange, blue, red and rose. Hence violet will be opposite to yellow.
- (a); Common faces with number 2 are in same positions. Hence when the digit 5 is on the bottom then 1 will on the upper surface.
- (c); The adjacent faces to the face which have 3 points are 2, 1, 4 and 6 points. Hence 5 points will be on the face opposite to the face which contains 3 points
- (a); Here one of the two common faces (5) is in the same position, the remaining face with the 4 dots will be opposite to face with dots 2.
- (c); When 3 points are at the bottom then 4 points will be at the top.
- (a); The letters of the adacent faces to the face with letter A, are B, F, C and E. Hence D is the letter of the face opposite to the face with letters (A).

19. (d); In given figure D have C, E, B as their adjacent remaining alphabet is A. So it is opposite of D.
20. (b); According to question figure. A is opposite to D, F is opposite to B and E is opposite of C. So only option B can follow the condition.
21. (d); After counting clockwise direction in both diagram 3, 5, 2 and 3, 1, 6 are given so opposite of 2 is 6.
22. (b); After counting clockwise direction in both diagram triangle

3	5	4
3	6	2

 so 1 is opposite to 3 triangle.
23. (a); In given four figures the face which contain 6 spots is adjacent of 3, 2, 4, 5, so remaining is 1. Which is opposite to 6.
24. (a); According to the figure 4 have 3, 6, 2, 5 as their adjacent so remaining digit is 1 which is opposite of 4.

(25 to 30)



$$n = \frac{\text{Given side}}{\text{Calculate}} = \frac{15 \text{ cm}}{5 \text{ cm}} = 3$$

$$\text{total no. of cubes} = (3)^3 = 27$$

$$3 \text{ face coloured} = 8$$

$$2 \text{ face coloured} = 12$$

$$1 \text{ face coloured} = 6$$

$$\text{Uncoloured} = (n - 2)^3 = 1$$

$$25. (b); \text{Coloured with only green colour} = 2.$$

$$26. (c); \text{Cubes that have at least two colours} = 12 + 8 = 20$$

$$27. (a); \text{Uncoloured cube} = 1.$$

$$28. (d); \text{Cube that have red colour} = 9 + 9 = 18.$$

$$29. (a); \text{Cube that have blue but not with red}$$

$$= \text{Only blue} + \text{blue with green}$$

$$= 2 + 4 = 6$$

$$30. (c); \text{Cubes that having only one face coloured}$$

$$= 6(n - 2)^3 = 6(3 - 2)^3 = 6$$